Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Backend Software Engineer with 2+ years building **Java** and Python services, **REST APIs**, and PostgreSQL systems on **AWS**, with **test-driven development**, event-driven design, Docker, and CI/CD in Agile teams.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Designed normalized **PostgreSQL** and Azure SQL schemas with row-level security for a multi-tenant platform serving 500+ users, optimizing indexes and query patterns to cut dashboard latency from 1.2s to 350ms.
- Built ETL-style integration services across FastAPI, PostgreSQL, SQL Server, and cloud storage with validation, reconciliation checks, and **CI/CD** releases, processing 5K+ monthly jobs at 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Built **REST APIs** behind a React dashboard that consume OpenAI, Anthropic, and Qwen **third-party APIs** through a switchable provider layer, serving 150+ instructors and 10K+ daily requests on AWS RDS.
- Shipped GitHub Actions CI/CD releases with fallback paths for third-party OCR timeouts and per-call token cost logging, keeping API responses stable under peak load while cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Wrote 20+ **unit and integration** test suites with PyTest and Mockito and automated nightly Jenkins runs across Windows, Linux, and macOS, collaborating with Microsoft and GitHub stakeholders in Agile sprints.
- Built object-oriented **microservice** modules with Singleton and Factory patterns integrating AWS Glue and S3, improving API response times 30% and pipeline efficiency 68% while cutting technical debt 20%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Built eight **Java Spring Boot** microservices on PostgreSQL for authentication, resource allocation, billing, leads, and notifications, pushing **event-driven** real-time updates and synchronizing 2M+ records across multiple business locations.
- Automated Jenkins CI/CD with unit, integration, and end-to-end tests, **Docker** packaging, and staged zero-downtime deployments, achieving 95%+ successful build validation while usage analytics cut resource wastage by 42%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Spring Boot booking, pricing, coupon, and **payment REST APIs** on PostgreSQL integrating third-party payment gateways, sustaining sub-200ms latency across 50K+ records for mobile and web clients.
- Automated idempotent payment-gateway webhooks for refunds, disputes, and status changes, cutting manual refund handling 70%, and translated client **requirements** into an end-to-end architecture with 12+ mapped integration points.

Skills

Languages & Backend: Java, Spring Boot, Python, SQL, TypeScript, JavaScript, REST APIs, Microservices, FastAPI, Flask
Data & Events: PostgreSQL, SQL Server, Azure SQL, MongoDB, Event Sourcing, Event-Driven Design, Domain-Driven Design
Testing & Practices: TDD, JUnit 5, Mockito, PyTest, Integration Testing, Code Reviews, Agile/Scrum, Git, Maven, System Design
Cloud & DevOps: AWS (ECS, S3, RDS, Glue), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD, CodePipeline, Linux
Integrations: Third-Party APIs, Payment Webhooks, Idempotency, HMAC Signing, Retries & Backoff

Projects

LoanLedger - Event-Sourced Loan Servicing | *Java 21, Spring Boot, PostgreSQL, JUnit 5, Event Sourcing*

- Built an **event-sourced** loan servicing ledger in Java 21, Spring Boot, and PostgreSQL with optimistic concurrency and exactly-once webhook handling, passing 5,500 randomized accounting-invariant tests and 3,000 duplicate deliveries without double-posting.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a **fintech** platform integrating the Plaid third-party API for bank linking, with a token-based firewall, real-time transaction updates, and Kubernetes CI/CD deployments supporting 50K+ users and 1M+ transactions.

Linguistix | *Python, Docker, AWS ECS Fargate, CodePipeline*

- Developed and deployed a containerized RESTful data processing API in Python on **AWS ECS Fargate** with CodePipeline CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability.